

CRBC- An automated approach for Handwriting OCR

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Abstract—In the realm of information management, the digitization of handwritten documents is pivotal. This research introduces an advanced Handwritten Optical Character Recognition (HOCR) model, leveraging Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory networks (BiLSTM), and the Connectionist Temporal Classification (CTC) loss function. Together Convolutional RNN-based Bi-LSTM CTC (CRBC), demonstrates a robust 94% accuracy, the model adapts seamlessly across various domains, presenting a scalable solution for enhanced handwritten document processing. This fusion of machine learning and natural language processing techniques contributes to improved efficiency in information management, with potential applications in diverse industries and fields.

Index Terms—Handwritten character recognition (HCR), Convolutional Neural Networks (CNN), Optical Character Recognition (OCR), Connectionist Temporal Classification (CTC) Loss, Recurrent Neural Networks (RNN), Bidirectional Long Short-Term Memory (LSTM)

I. INTRODUCTION

The realm of technology has witnessed a surge in the adoption of digital solutions to streamline operations and data management. However, the prevalent use of handwritten and printed documents poses challenges [1] in terms of efficiency, accuracy, and accessibility. This research delves into the domain of document digitization, employing state-of-the-art deep-learning techniques to transform diverse document formats into machine-readable text. The proposed methodology harnesses the power of Optical Character Recognition (OCR) and advanced neural network architectures to automate the transcription process, enhancing the accessibility and usability of document data for both individuals and organizations [2].

Recent advancements in OCR technology have paved the way for significant progress in the field of handwritten character recognition (HCR). Convolutional Neural Networks (CNNs), with their exceptional ability to extract spatial features, have emerged as a prominent deep learning architecture for HCR tasks [3]. Researchers have further explored recurrent neural networks (RNNs) and Bidirectional Long Short-Term Memory (LSTM) layers to capture sequential dependencies in handwritten text, improving recognition accuracy. The Connectionist Temporal Classification (CTC) loss function has also gained popularity due to its effectiveness in handling unsegmented sequences, a common challenge in HCR.

The primary challenge addressed in this research lies in the digitization of handwritten and printed documents. The

intricate variations in handwriting styles, diverse document formats, and the presence of abbreviations and specialized terminology pose significant hurdles for existing OCR systems. The lack of standardized metrics and cross-dataset validation further complicates the evaluation and comparison of different HCR models [4].

To address the aforementioned challenges, this research proposes a methodology for prescription digitization that integrates advanced deep-learning techniques. The proposed model combines CNNs for spatial feature extraction, RNNs, and LSTM layers for sequential modeling, and the CTC loss function for unsegmented sequence handling to generate a robust system called Convolutional RNN-based Bi-LSTM CTC (CRBC) model. By leveraging these state-of-the-art approaches, the model aims to achieve accurate and robust transcription of prescription data, regardless of the handwriting style or prescription format.

Our contribution is to divide the handwritten and printed text work on them independently and propose a suitable method for text extraction. The proposed methodology integrates advanced deep learning techniques, specifically Convolutional Neural Networks (CNNs) and Bidirectional Long Short-Term Memory (BiLSTM) layers, to address the challenges of handwritten optical character recognition (OCR). Our goal was to create a reliable and strong text recognition system while optimizing accuracy, speed, and flexibility. We investigated three different model training strategies, each utilizing cutting-edge deep learning architectures, to do this. Additionally, the incorporation of the Connectionist Temporal Classification (CTC) loss function enhances the model's ability to handle unsegmented sequences, a common scenario in handwritten OCR.

The paper is structured as follows:

In Section II, the literature survey reviews existing research relevant to the topic under investigation, providing an overview of the current state of knowledge in the field. Section III details the proposed methodology and Convolutional RNN-based Bi-LSTM CTC (CRBC), accompanied by background knowledge and illustrative architectural and process flow diagrams. Following this, Section IV presents the experimental results obtained from the evaluation of the CRBC model. This section scrutinizes the performance of the algorithm and presents findings derived from the experimental analysis. Finally, Section V serves as the conclusion section, summarizing the key findings

of the research.

II. LITERATURE SURVEY

Handwritten Optical Character Recognition (HOCR) serves as a pivotal technology, facilitating the conversion of diverse document formats into editable and analyzable data. Its primary goal is to emulate human reading capabilities, enabling machines to read, edit, and interact with text akin to humans. Over the past five decades, HOCR has undergone significant evolution, drawing considerable attention from researchers and yielding notable advancements [5].

Recent progress in technology has witnessed a shift towards utilizing vision sensors for handwriting recognition, capturing three-dimensional finger movements during writing. This has opened avenues for enhancing the accuracy and efficiency of HOCR systems. Singh et al. [6] presented a Neural Network-based HOCR model that outperformed commercial Handwritten Text Recognition (HTR) cloud APIs, achieving an impressive error rate of 7.6%. One notable application of HOCR is in the digitization of regional languages, enabling the preservation and analysis of diverse linguistic scripts. Rahman et al. demonstrated the effectiveness of handwritten OCR for Bengali texts, showcasing its proficiency in preserving and analyzing handwritten documents in the Bengali script [7]. Similarly, in the realm of the Hebrew language, Suissa et al. highlighted the application of OCR for accurate transcription and digitization of handwritten Hebrew manuscripts [8].

Despite these advancements, challenges persist due to diverse handwriting styles, similar characters, and a broad range of character categories. In response, convolutional neural networks (CNNs) have emerged as a prominent deep learning architecture [9], inspired by human visual perception processes. Abbas et al. proposed a CNN-based system that achieves up to 93% accuracy in recognizing handwritten documents, offering promising support for visually impaired users and automatic data entry [10]. The fundamental components of CNNs, including fully connected, pooling, and convolutional layers, have significantly enhanced HOCR performance, driving progress in the broader field of computer vision [11].

Expanding on deep learning approaches, Khandokar et al. [12] demonstrated the proficiency of CNNs in HOCR, achieving an accuracy of 92.91%. Additionally, Adak et al. proposed CNN-based architectures with competitive Character Error Rates (CER) of 6.21%, respectively, underlining the effectiveness of CNNs in addressing HOCR challenges [13].

Moving beyond traditional CNNs, Li et al. introduced TrOCR, a Transformer-based approach that showcased state-of-the-art results in various text recognition tasks [14], [15]. In parallel, Guan et al. utilized a style-conditioned generative adversarial network (SC-GAN) to transfer styles from real handwriting to skeleton images, generating photo-realistic text line images [16]. Complementarily, Vögtlin, et al. employed GANs in their work on generating synthetic handwritten historical documents with OCR-constrained GANs, contributing to the generation of large labeled historical document datasets with precise ground truth [17].

Moreover, advancements in HOCR continue with the incorporation of novel models for generating training samples. Ding et al. introduced a denoising diffusion probabilistic model (DDPM), conditioning printed glyph images to create mappings between printed characters and handwritten images. This innovation enables the generation of photo-realistic handwritten samples with diverse styles and unseen text contents, showcasing the continuous evolution of HOCR techniques [18]. The intersection of regional languages and HOCR is exemplified by Balaha's HMB-AHCR-DLGA approach, combining transfer learning and genetic algorithms to achieve a testing accuracy of 92.88% for Arabic handwritten character recognition [19].

Furthermore, the literature encompasses a comparative study by Liang on the layout analysis of tabulated historical documents, shedding light on challenges and advancements in handling complex document layouts [20]. Geetha et al. presented a hybrid handwritten text recognition model using CNN-RNN networks and a sequence-to-sequence approach, showcasing competitive letter and word accuracy in recognizing handwritten text [21]. In the medical field, Karthikeyan's work introduces a deep neural network-based method that significantly reduces word error rates in OCR of medical reports, demonstrating the versatility of OCR applications in specialized domains [22].

Despite these remarkable advancements, the literature underscores the importance of standardizing metrics, conducting cross-dataset validations, and exploring the scalability of models to meet the evolving demands of handwritten character recognition. The incorporation of diverse models and methodologies reflects the dynamic landscape and continuous progress in the field of HOCR.

III. CRBC METHODOLOGY

In this section, we introduce the Convolutional RNN-based Bi-LSTM CTC (CRBC), our proposed methodology for high-precision text extraction. We delve into the background, architectural design, and process flow of CRBC, providing a comprehensive understanding of its implementation and functionality.

A. Background

Deep learning has revolutionized handwritten optical character recognition (HCR), with models like Convolutional Neural Networks (CNNs), Bidirectional Long Short-Term Memory (BiLSTMs), and the Connectionist Temporal Classification (CTC) loss function enabling exceptional accuracy. These models form the foundation of CRBC, and their mathematical underpinnings are essential for understanding the methodology's efficacy.

1) Convolutional Neural Networks (CNNs)

CNNs excel at image classification due to their ability to extract hierarchical features. We leverage a sequence of convolutional layers to capture patterns at various scales within input images, crucial for accurate text recognition.

a) *Mathematical Representation of Layer Operations*

Let input data be denoted as X , with layer parameters W (weights) and b (biases).

Neurons, Weights, and Biases: Each neuron in a convolutional layer corresponds to a receptive field in the input image, learning to detect patterns within this region. Weights enable feature detection and are shared across regions, while biases introduce flexibility in shifting activation functions.

Convolution Operation: The convolution operation extracts features by sliding a convolutional kernel over the input, performing element-wise multiplication and summation:

$$\text{Convolution}(P, W, B) = P * W + B \quad (1)$$

where: P is the input tensor, W is the convolutional kernel (weights), B is the bias term, and $*$ denotes the convolution operation. The convolution operation involves sliding the convolutional kernel over the input, element-wise multiplication, and summation to produce a feature map. The result is a feature map that highlights relevant patterns in the input data.

MaxPooling Layer: Max pooling downsamples input tensors by retaining the maximum value within defined regions:

$$\text{MaxPooling}(Y) = \max(Y) \quad (2)$$

where $\max(Y)$ returns the maximum value within the specified pool size.

Dropout Layer: Dropout mitigates overfitting by randomly dropping units during training:

$$\text{Dropout}(Y, \text{rate}) = Y \odot \text{dropout_mask} \quad (3)$$

where the dropout mask is generated based on the dropout rate.

Softmax Activation Layer: The softmax function computes probabilities for multi-class classification, normalizing exponentials of input vector elements:

$$\text{Softmax}(Y) = \frac{e^{Y_i}}{\sum_j e^{Y_j}} \quad (4)$$

ReLU Activation Layer: Rectified Linear Unit (ReLU) is a common activation function that introduces non-linearity. For input Y , $\text{ReLU}(Y)$ returns the element-wise maximum of X and zero, promoting sparse activation and efficient learning.

$$\text{ReLU}(y) = \max(0, y) \quad (5)$$

where y is the input to the ReLU function. The function returns the input value if it is positive; otherwise, it returns zero.

2) *Bidirectional Long Short-Term Memory (BiLSTM) Layers*

Sequential modeling was introduced by integrating one RNN layer into our hybrid architecture. Because RNNs are so good at identifying relationships in sequential data, they are a great tool for applications requiring ordered information,

including text recognition. We sought to improve our model's understanding of textual elements' sequential character by incorporating RNNs, which would enable a more thorough understanding of intricate linguistic patterns. We implemented

Bidirectional Long Short-Term Memory (BiLSTM) networks to strengthen our model's understanding of sequential dependencies. Long-range dependencies are better captured by the model because of the bidirectional nature of LSTMs, which allows the model to take into account both past and future context. This is especially useful in situations when a character's context is impacted by both characters that come before and after it, which is frequently the case in textual data.

Bidirectional LSTMs, represented by `lstm1` and `lstm2` layers, process sequences bidirectionally, capturing dependencies efficiently. The forward and backward hidden states $\vec{h}_1$, and $\overleftarrow{h}_1$, are concatenated to provide a comprehensive understanding of temporal relationships in the data. Each BiLSTM layer contains 256 units, enhancing the model's capacity to unravel intricate linguistic patterns in sequential data.

3) *Connectionist Temporal Classification (CTC) Loss Function*

A key element of our methodology was the addition of the Connectionist Temporal Classification (CTC) loss function. By resolving alignment problems between input images and matching textual sequences, CTC makes end-to-end training easier. As a result, less explicitly aligned data were required, resulting in a more adaptable and scalable training procedure. With the use of CTC, our model was able to bridge the gap between visual features and their linguistic representations by learning to align and recognize characters within the input images.

In the context of neural network training for sequence-to-sequence tasks, the Connectionist Temporal Classification (CTC) loss function plays a crucial role. The CTC loss is particularly useful in scenarios where the alignment between input and output sequences is not explicitly known during training, making it suitable for tasks like handwritten text recognition.

The CTC loss function is expressed mathematically as:

$$\text{CTC}(y, \pi) = -\log \sum_{\pi' \in B^{-1}(\pi)} \prod_{t=1}^T P(y_t | x, \pi'_t) \quad (6)$$

Here, y represents the true label sequence, π denotes a possible alignment between the input and output sequences, $B^{-1}(\pi)$ is the set of all possible label sequences mapping to π , T is the length of the input sequence, and $P(y_t | x, \pi'_t)$ is the predicted probability of the t -th character y_t given the input x and alignment π'_t . The summation is taken over all possible alignments π' , and $\log$ refers to the natural logarithm.

The CTC loss function, in essence, is designed to optimize the likelihood of the correct label sequence by considering all potential alignments of the input data. During training, the model learns to assign higher probabilities to correct label sequences while penalizing incorrect alignments. This

flexibility in handling variable-length mappings is particularly advantageous in sequence-based tasks such as handwritten text recognition.

B. CRBC Architecture

CRBC leverages the strengths of CNNs, BiLSTMs, and CTC to achieve high-precision text extraction. Figure 1 illustrates the architecture, highlighting the interplay between different layers.

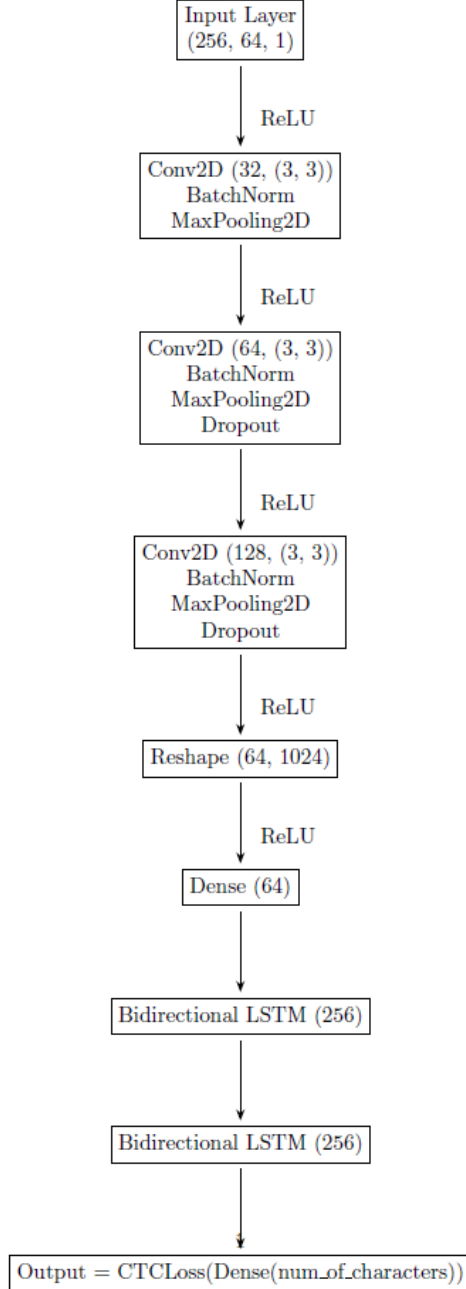


Fig. 1. Proposed Model Architecture representing each layer and its weights and shapes, also underlying the Convolution layers, Bi-Directional LSTM layer, and CTC Loss function

The architecture begins with an input layer (`input_data`) designed to accept grayscale images of dimensions 256x64x1,

a standard format for representing handwritten characters. Following the input layer are three convolutional layers (`conv1`, `conv2`, `conv3`), each playing a crucial role in feature extraction. These layers employ a combination of techniques to progressively extract meaningful features from the input images.

Conv2D Layers: Each convolutional layer utilizes a set of learnable filters to detect specific patterns within the input image. The first layer, `conv1`, starts with 32 filters, while the subsequent layers, `conv2` and `conv3`, gradually increase the number of filters to 64 and 128 respectively. This hierarchical approach allows the network to learn increasingly complex and abstract representations of the input data.

Kernel Size and Padding: A kernel size of (3, 3) is used in all three convolutional layers. This size has proven effective in capturing local patterns within images. 'Same' padding is employed to ensure that the output feature maps maintain the same spatial dimensions as the input, preventing the loss of information at the edges.

Batch Normalization: After each convolutional layer, Batch Normalization is applied to standardize the activations. This normalization process helps stabilize training by reducing internal covariate shift, leading to faster convergence and improved generalization.

ReLU Activation: The Rectified Linear Unit (ReLU) activation function introduces non-linearity into the network. ReLU's simplicity and effectiveness in addressing the vanishing gradient problem make it a popular choice for deep learning architectures.

MaxPooling2D: MaxPooling2D layers with varying pool sizes are strategically placed after each convolutional layer. The first two MaxPooling layers use a pool size of (2, 2), effectively downsampling the spatial dimensions of the feature maps by a factor of 2 in both height and width. The third MaxPooling layer uses a pool size of (1, 2), performing downsampling only along the width dimension. This selective downsampling helps to make the model more robust to variations in handwriting styles and character positions.

Dropout Layers: To mitigate the risk of overfitting, two dropout layers are incorporated after the second and third convolutional layers. Both dropout layers utilize a dropout rate of 30%, randomly deactivating a fraction of the neurons during training. This regularization technique prevents the network from becoming overly reliant on any particular set of features, promoting better generalization to unseen data.

The transition from convolutional layers to recurrent layers requires reshaping the output from the last convolutional layer. The Reshape layer transforms the multi-dimensional feature maps into a format suitable for sequential processing by the subsequent BiLSTM layers.

A dense layer (`dense1`) with 64 units and ReLU activation follows the reshape operation. This layer acts as a bridge between the convolutional and recurrent parts of the network,

allowing for a smoother integration of spatial and sequential information.

The heart of CRBC's sequential processing capability lies in its two Bidirectional Long Short-Term Memory (BiLSTM) layers (`lstm1` and `lstm2`). Each BiLSTM layer contains 256 units, enabling the network to capture long-range dependencies within the sequence of features extracted by the convolutional layers. These layers process the input sequence in both forward and backward directions, capturing contextual information from both preceding and succeeding elements.

The output from the final BiLSTM layer is then passed through a dense layer (`dense2`) responsible for character prediction. The number of units in this layer corresponds to the number of distinct characters in the dataset, allowing the network to output a probability distribution over all possible characters for each position in the sequence.

Finally, the softmax activation function is applied to the output of the dense layer to obtain a probability distribution over the character vocabulary. This distribution represents the network's confidence in its predictions for each character in the sequence.

The integration of the Connectionist Temporal Classification (CTC) loss function is paramount to CRBC's effectiveness in handwritten text recognition. CTC operates by considering all possible alignments between the input image and the output character sequence, eliminating the need for pre-segmented characters during training. This characteristic is particularly valuable for handwritten text, where character boundaries can be ambiguous. By evaluating multiple alignments, CTC guides the network to learn the most probable character sequence even from unsegmented input.

IV. EXPERIMENTAL RESULTS

In this section, a comprehensive analysis of the experimental results derived from the proposed methodology applied to the chosen dataset is presented. This investigation covers key performance metrics such as accuracy and CTC loss function, along with model evaluations conducted through rigorous testing on distinct subsets—specifically, the training, testing, and validation sets. The experimental findings illuminate the effectiveness of our approach in character recognition and offer valuable insights into the model's ability to accommodate diverse writing styles present in the dataset.

A. Dataset

The dataset¹, encompassing over four hundred thousand handwritten names, serves as a comprehensive resource gathered through charitable initiatives. Focused on character recognition, the dataset employs image processing technologies to convert characters from scanned documents into digital formats. While machine-printed fonts typically yield satisfactory results, recognizing handwritten characters presents intricate challenges due to the significant variation in individual writing styles.

¹Handwriting Dataset used from Kaggle: <https://www.kaggle.com/datasets/landlord/handwriting-recognition>

Each image is meticulously labeled, following a standardized naming format, facilitating seamless integration with additional data. This rich dataset forms the backbone of our research, providing a diverse and extensive collection of handwritten names for the development and evaluation of advanced character recognition models.

B. Results

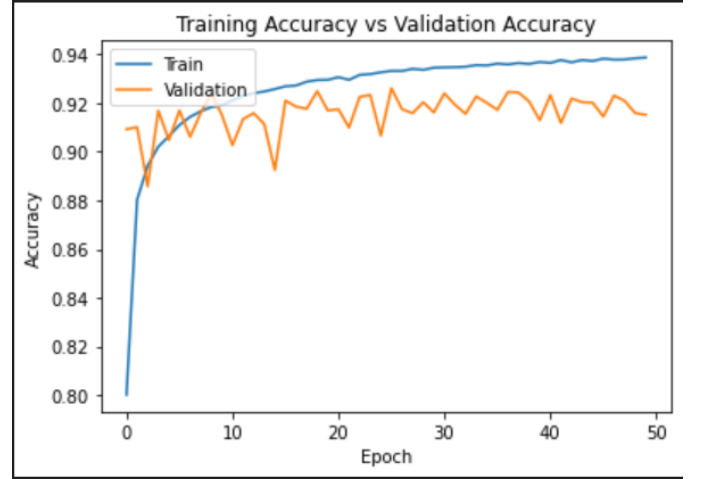


Fig. 2. Training Vs Validation Accuracy

The training and testing were conducted over 50 epochs using the Adam Optimizer with a learning rate of 0.0001 and a batch size of 120. The experiments were performed on Python 3.10, utilizing a computer with the following specifications: an i5 12600K CPU, 16GB of RAM, and 500GB of SSD storage, from Figure. 2, the training accuracy stabilizes in the final epochs, indicating potential convergence but the validation accuracy fluctuates, suggesting potential overfitting concerns.

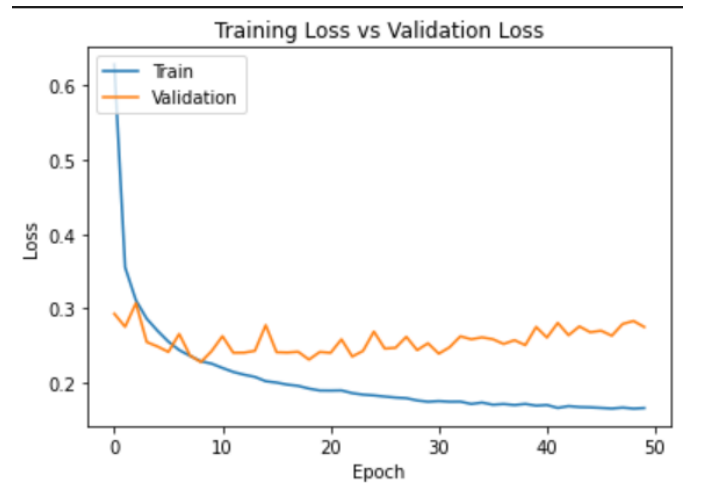


Fig. 3. Training Vs Validation Loss

Similarly, in the Figure. 3 training loss demonstrates rapid decrease, contrasting with the validation loss which exhibits

minor fluctuations but increases slightly towards the end, indicating a possible loss of generalization.

V. CONCLUSION

In conclusion, this research has explored the complex field of handwritten character recognition (HCR) and the various advancements achieved therein. The reviewed literature encompasses a range of methodologies, including Convolutional Neural Networks (CNN) and Transformer-based models, which collectively shape the evolving landscape of HCR. Of particular note is the proposed hybrid model integrating CNN, RNN, Bidirectional LSTM, and CTC Loss, designed to tackle challenges in digitizing prescriptions. Experimental findings on a diverse dataset have shown promising outcomes, achieving accuracy rates of up to 94%. Despite significant progress, there remains a need for standardizing metrics and validating results across different datasets. This study underscores the potential benefits of technology in enhancing efficiency and data accessibility while facilitating the adoption of advanced machine-learning techniques across various sectors.

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